

Placing scripted events

It is the job of level designers to introduce scripted event locations, where certain actions by the player can trigger specified changes in the level environment. Scripted events are any events programmed into a level by the level designer to make any number of things happen. They can be as simple as a player reaching a closet and an enemy suddenly comes out of it or as complex as changing the level exits. Generally they are used to move a story forward as the player to do something specific to keep the story in control. Cutscenes can be considered a scripted event when they are rendered in-engine like fight between two NPCs and player just has to watch it to remain in context with the story. Typical RPG quest lines and missions can also be considered a long chain of scripted events.

More often in single player games, they are used to complement AI by relatively complex actions being scripted. Though it is always suggested to let AI and physics engine handle such events so as to keep the world dynamic and interactive.

Placing path finding nodes

Though complex path finding solutions are implemented by AI systems discussed later, it is often suggested that level designers can place path-finding nodes that non-player characters take as they walk around to test level navigation and design. Using popular level editors, it is easy to place path finding nodes in a level.

Deciding the level aesthetic

Later stages in level design include adding aesthetic details to the level such as music, textures animation and effects.

Tools used

- ▶ **Graphic design software:** Adobe Photoshop, Illustrator, GIMP.
- ▶ **Level editors:** Valve's Hammer Editor, Epic's UnrealEd, Leadwerks 3D World Studio, BioWare's Aurora Toolset, id Software's Q3Radiant, Unity 3D and Grome outdoor editor. Multi engine, multi game editors include id Software's GtkRadiant, based on Q3Radiant, and the open source Quark.
- ▶ **Games with inbuilt level editors:** *Battlezone 2*, *Cube 2: Sauerbraten* and *Doom 3*, *TimeSplitters*
- ▶ **Professional 3D editing software:** 3D Studio Max, Blender, AutoCAD, Lightwave, Maya, Softimage XSI or Grome.
- ▶ **Terrain Generator Software:** Terragen, Bryce, Vue, XFrog.